

Computer Networks

Lecture 2: Introduction

Content



- **Types of Computer Network**
- **Network Topology**

Comparison between Simplex, Half-duplex and Full-duplex

Simplex	Half-duplex	Full-duplex
Unidirectional communication.	Bi-directional communication but alternately.	Bi-directional communication and simultaneously.
Sender can only send the data but can't receive the data, similarly receiver can only receive the data.	Both communicating nodes can send and receive the data alternately.	Both communicating nodes can send and receive the data simultaneously.
Performance is less than other two modes of communication.	It provides less performance than full duplex	It provides the best performance among three modes of communication.
Example: Keyboard and monitor.	Example: Walkie-talkie.	Example: Telephone, mobile.

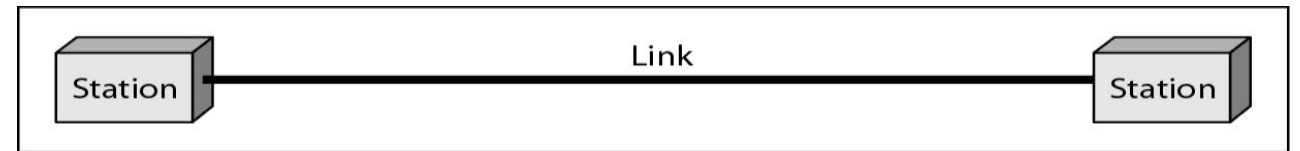
Line configuration

Definition

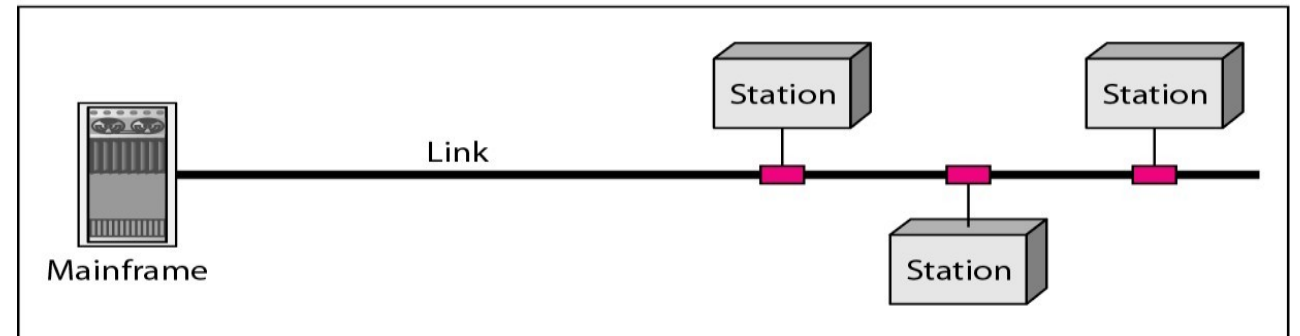
Line configuration, also known as connection configuration, defines how devices are connected to a transmission medium or communication link.

There are two possible types of connections:

- Point to Point
- Multipoint



a. Point-to-point



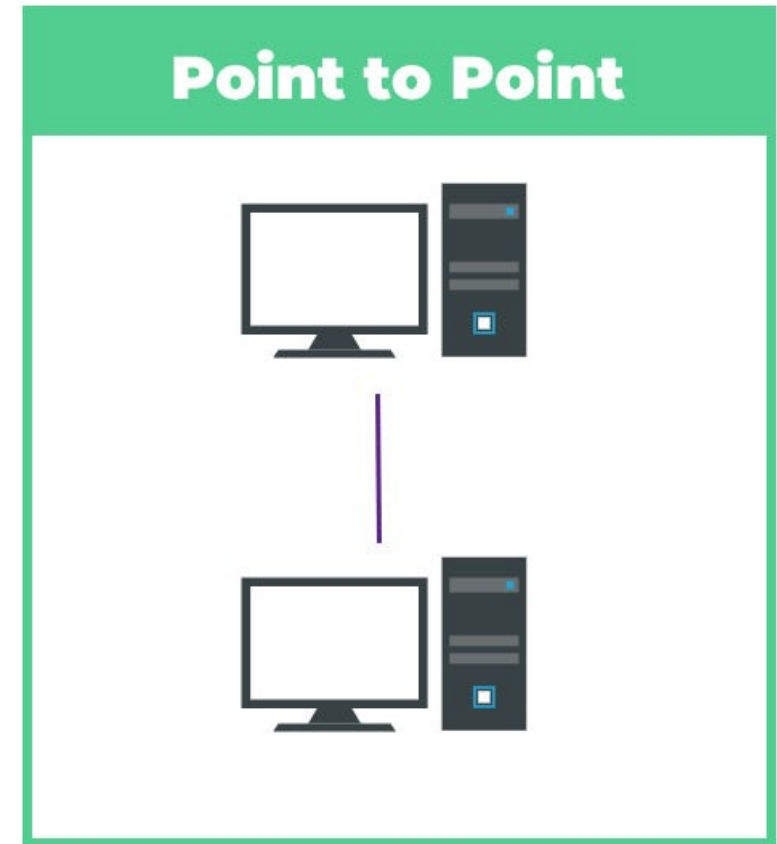
b. Multipoint

Point-to-point connection

Provides a dedicated link between precisely two devices. The full capacity of the link is reserved for communication between these devices.

Advantages

- Simple setup
- Maximum bandwidth availability
- More secure as communication is private
- Better performance due to dedicated resources

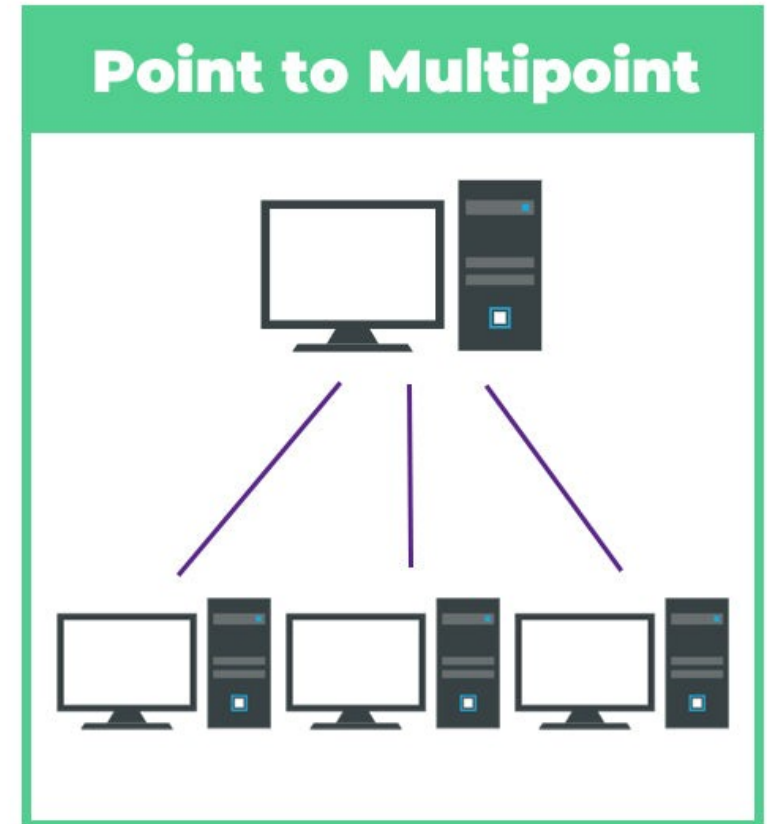


Multipoint connection

Connects three or more devices using a single shared link. All devices share the channel capacity

Characteristics:

- More complex than point-to-point
- Cost-effective for larger networks
- Requires access control mechanisms
- Bandwidth is shared among all devices



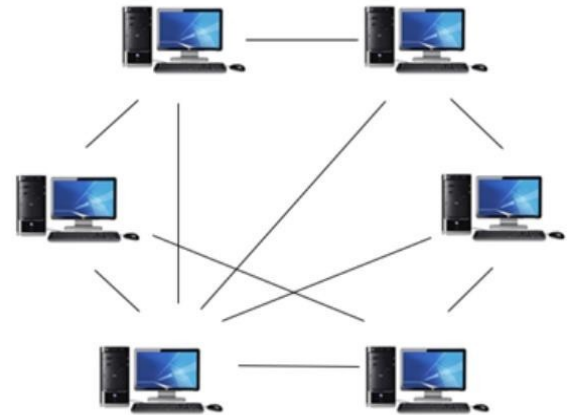
Computer network architectures

Network architecture refers to the physical and logical structure that defines how software, hardware, protocols, and transmission mediums work together to facilitate data transmission across a network.

- There are two most popular network architectures:
 - Peer-to-Peer architecture
 - Client-Server architecture

Peer to Peer Architecture

In a peer-to-peer (P2P) network architecture, all computers are connected to each other, and each computer has equal privileges and responsibilities for sending, receiving, and processing data.



- **Advantages:**
 - Cost-effective
 - Easy to configure
 - No single point of failure: If one computer goes down, others can continue to function in the network.
- **Disadvantages:**
 - Security risk is high
 - Data backup is not easy

Client-Server architecture

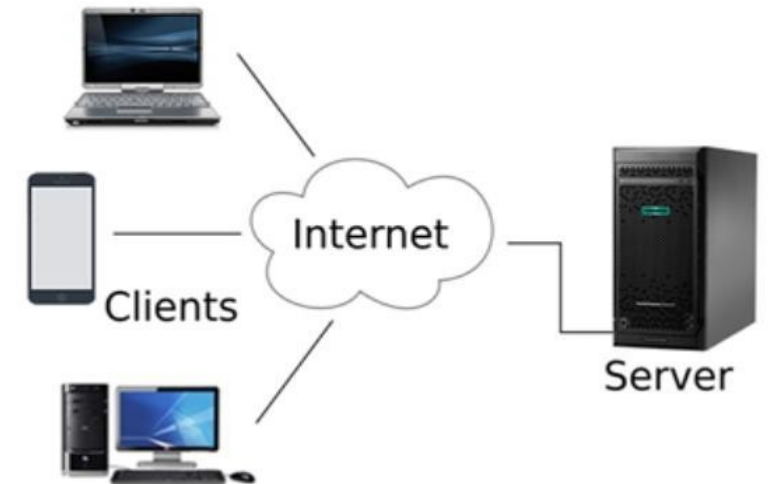
In a client-server architecture, all computers are connected to a central system called a server. All communication in the network is managed through this server.

Advantages:

- Data backup is easy
- Improves the overall performance of the network
- Better security than peer-to-peer network

Disadvantages:

- It is expensive.
- It requires a dedicated network administrator to manage all the resources.
- Single point of failure: If the central server fails, it can disrupt the entire network.



Types of computer networks

The type of a network refers to the scale and scope of the network based on its geographical range and the number of devices it connects.

Common types include:

- Personal Area Network (PAN)
- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)

PAN

Purpose

PAN is used to connect personal devices for individual use.

Components: Typically includes devices such as mobile phones, computers, and personal digital assistants.

Range: Generally covers a distance of up to 10 meters.

Environment: Provides a single-user network environment to connect multiple devices.

Technology: Can use either wired connections (e.g., USB) or wireless technologies (e.g., Bluetooth, Wi-Fi).

Cost and Coverage: A low-cost network designed for small-area communication, primarily for personal devices.

LAN

Purpose

LAN is used to connect devices within a limited geographic area, such as a home, office, or campus.

Components: Includes devices such as computers, printers, servers, and network switches.

Range: Generally covers a distance of up to one kilometer.

Environment: Supports multiple users and devices within a confined area, providing high-speed connections.

Technology: Commonly uses wired connections (e.g., Ethernet) and wireless technologies (e.g., Wi-Fi).

Cost and Coverage: A relatively low-cost network ideal for small to medium-sized areas with high-speed connectivity.

MAN

Purpose

MAN is used to connect networks across a city or large campus, providing high-speed data transfer over a larger area than a LAN.

Components: Includes routers, switches, and fiber-optic cables, linking multiple LANs within a metropolitan area.

Range: Typically covers a distance of up to 50 kilometers.

Environment: Supports medium to large organizations or cities, facilitating communication across various locations.

Technology: Utilizes high-speed connections, often involving fiber optics or leased lines.

Cost and Coverage: More costly than LANs, designed for wide-area coverage within a metropolitan region.

WAN

Purpose

WAN is used to connect devices and networks over large geographic areas, including different cities, countries, or even continents.

Components: Involves a variety of networking equipment such as routers, switches, and satellite links.

Range: Can span thousands of kilometers, covering global distances.

Environment: Supports large-scale networks for organizations with multiple branches or remote locations.

Technology: Employs various technologies including leased lines, satellite communications, and VPNs (Virtual Private Networks).

Cost and Coverage: More expensive due to the extensive infrastructure and global reach required.

Network topology

Network topology is the physical and logical arrangement of devices in a network. The following are the major network topologies possible in the computer networks.

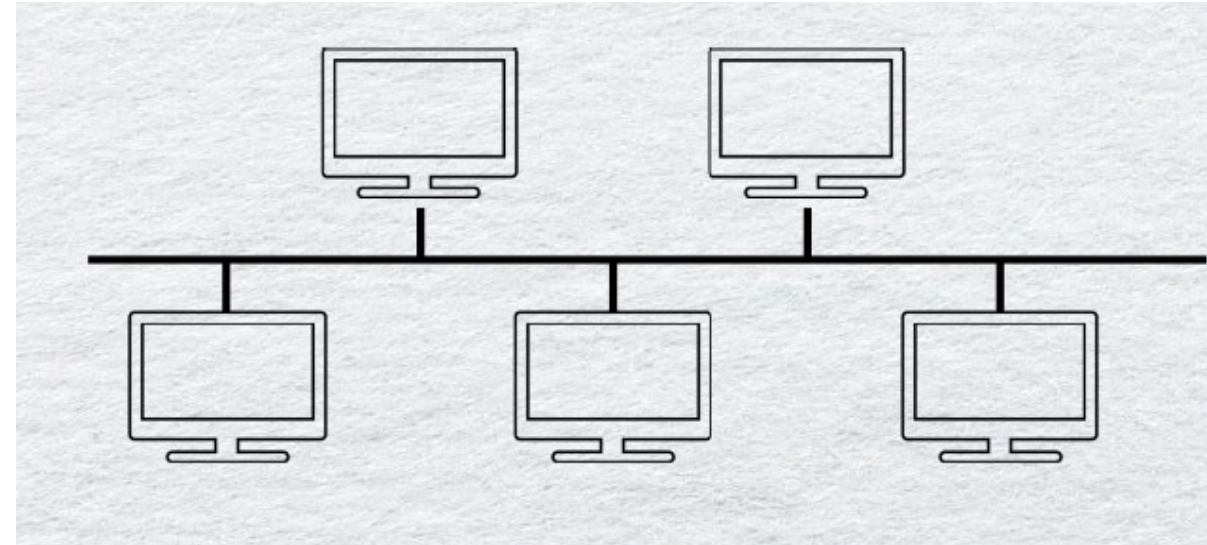
- ✓ Bus Topology
- ✓ Star Topology
- ✓ Mesh Topology
- ✓ Ring Topology
- ✓ Tree Topology
- ✓ Hybrid Topology

The choice of network topology significantly impacts how a network functions.

Bus topology

In a bus topology, there is a backbone cable called a bus, and all the devices are connected to this bus through drop lines and connectors.

- ✓ The drop line is a connection cable between a device and the bus
- ✓ Terminators are installed at both physical ends of the bus. They absorb the signals when they reach either end.
- ✓ Communication is half duplex. It means, only one device can send the data over the shared bus at any given time.



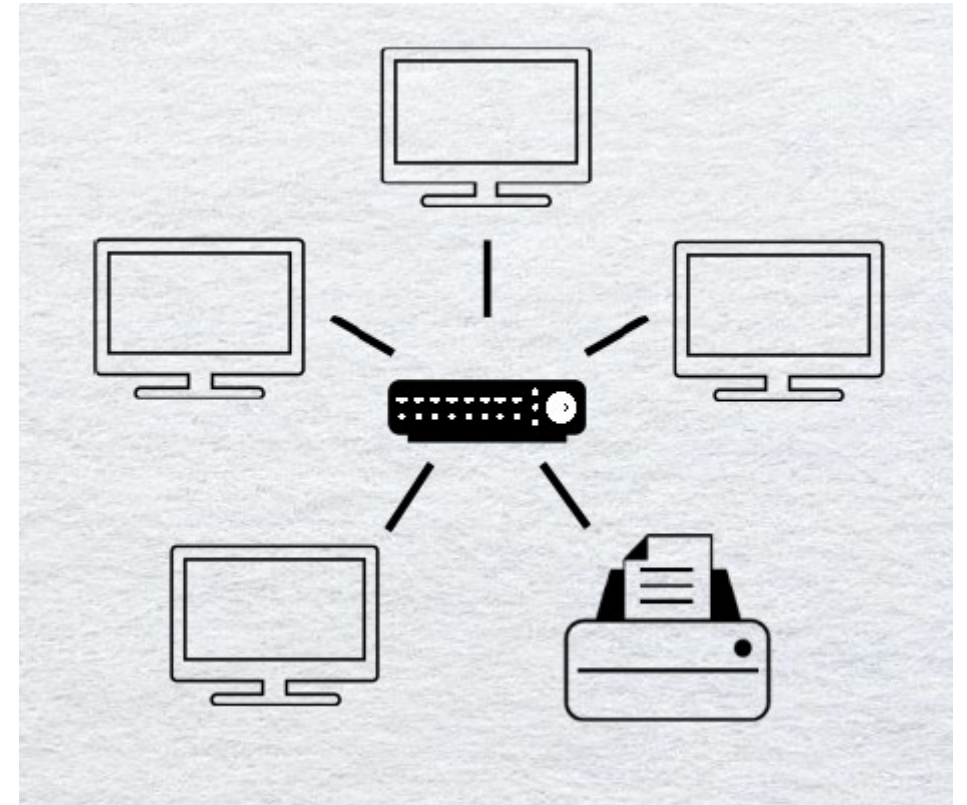
Bus topology

Advantage	Disadvantage
It is easy to install and inexpensive to implement	Difficult to troubleshoot
It requires fewer cables as compared to other topologies	Limited scalability
A single device failure does not impact the operation of other devices.	It is not fault-tolerant
Well suited for temporary networks	Security is very low

Star topology

In a star topology, each device in the network is connected to a central device called a hub or switch.

- ✓ In this topology, the devices cannot communicate with each other directly, they can communicate with other devices through a switch or hub.
- ✓ In star topology, communication is full-duplex. It means, devices can send and receive data simultaneously



Star topology

Advantage	Disadvantage
It is robust	If the central switch or hub fails, the entire network becomes inoperative
Fault detection is easy	High Maintenance cost for hub or Switch
It is scalable	
Well suited for temporary networks	

Mesh topology

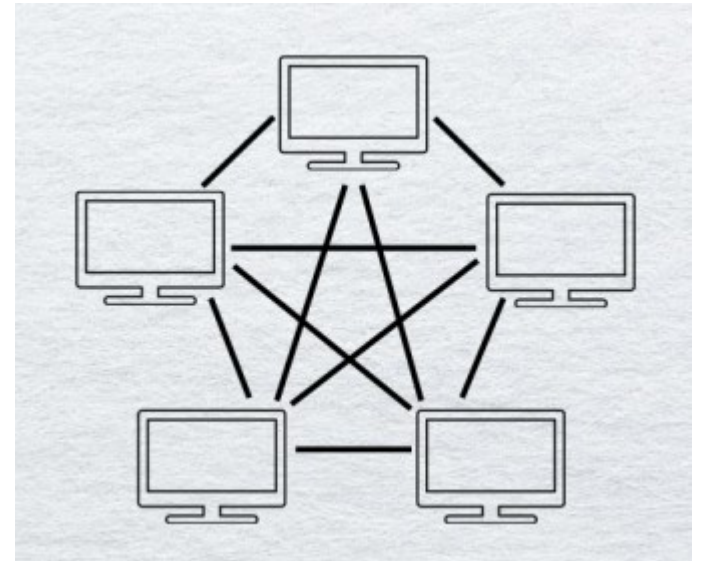
In a mesh topology, each device is connected to other devices in the network through a dedicated point-to-point link. There are two types of mesh topologies:

- ✓ Fully connected mesh topology
- ✓ Partially connected mesh topology

In a fully connected mesh topology, each device is connected to all other devices in the network through dedicated point-to-point links.

If there are n devices in the network, then links are possible in a fully connected mesh topology.

In a partially connected mesh topology, not all devices need to be connected to each other.



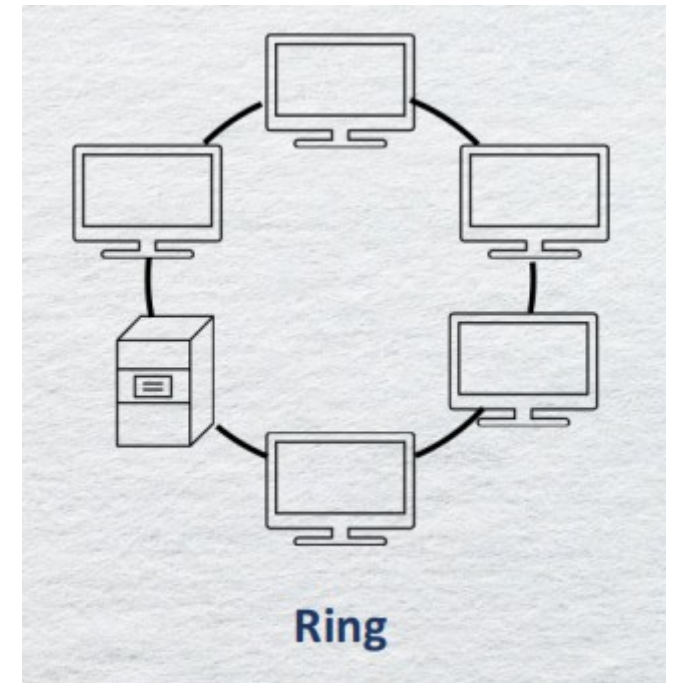
Mesh topology

Advantage	Disadvantage
No data traffic issues	Mesh topology can be expensive due to the extensive wiring need
It is reliable and robust	More maintenance is required in mesh topology
It is secure	
Fault in the network is diagnosed easily	

Ring topology

In a ring topology, devices are connected in a circular loop, and each device is connected to exactly two other devices. There are two major types of ring topologies

- ✓ Unidirectional ring: In this topology, data travels in one direction around the ring, passing through each device until it reaches its destination.
- ✓ Dual or bidirectional ring: In this topology, communication is full-duplex, meaning data can travel in both directions simultaneously, improving efficiency and fault tolerance.



Note: It works based on a token system, where the device with the token can send data to another device.

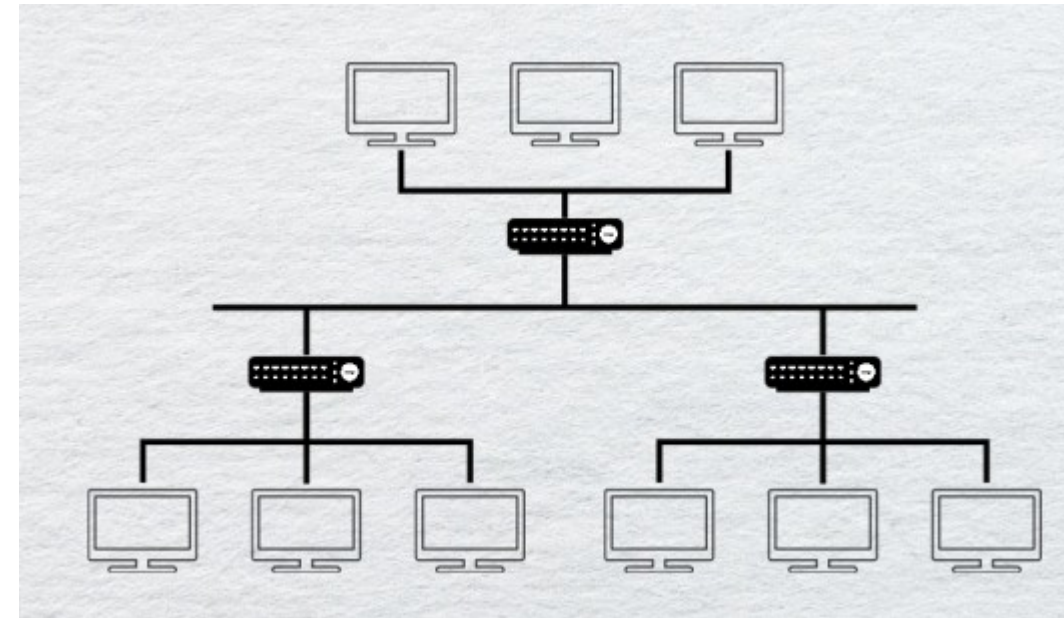
Ring topology

Advantage	Disadvantage
No central node is required	If a node goes down entire network goes down.
Each device gets equal access to the network	Data must pass through each node to reach its destination.
The speed of data transfer is very high in this type of topology	Higher initial setup costs compared to bus or star topologies.
Due to the presence of token passing, the performance of ring topology becomes better than that of bus topology under heavy traffic	

Tree topology

It combines bus and star topology characteristics. In tree topology, all the devices are connected hierarchically, with a branching structure resembling a tree.

- ✓ There is a central root device, often a hub or switch, from which all other devices branch out.
- ✓ Tree topologies are commonly used in large corporate networks and data centers due to their scalability and hierarchical structure.



Tree topology

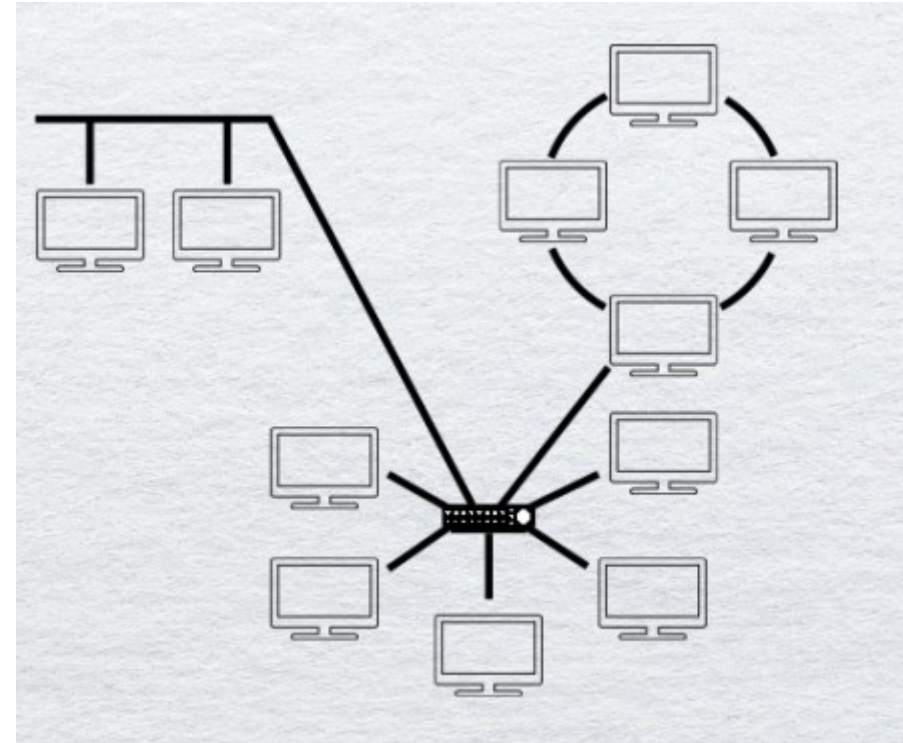
Advantage	Disadvantage
Scalability	Tree topology can be expensive to implement because it requires a large number of cables and network devices.
Better fault tolerance	Single point of failure: If the top-level node fails, it can impact the entire network.

Hybrid topology

Hybrid topology is a combination of two or more topologies:

Examples:

- ✓ Star-Bus topology,
- ✓ Bus-Star-Ring topology,
- ✓ Ring-Star topology



Tree topology

Advantage	Disadvantage
Customizable design	Design is complex so maintenance is high.
Scalability	Installation is difficult.
	Fault detection is difficult

Questions

1. In a bus topology, if $N = 15$ devices are connected to a single backbone cable, how many drop lines are required?
2. A star network has 20 computers connected to a central switch. Find the total number of links.
3. If 12 nodes are connected in a ring topology, how many communication links are required?
4. A fully connected mesh topology contains 8 nodes.
 1. Calculate the number of links required.
 2. Degree of each node
5. A hybrid network consists of 5 star networks, each with 6 nodes, connected using a bus backbone. Find:
 1. Total number of nodes
 2. Total number of links inside star networks.